

Oyster GameData Recorder — PRD Requirements vs Current State (Gap Analysis)

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PRD Requirements vs Current State — Gap Analysis

Date: 2026-05-12 **Reference PRD:** `docs/PRD.md` v1.0 + `bin/lint_v3_prd_grounded.py` 32 acceptance criteria **Shipped:** rc18.0.5 (live), rc18.0.6 (CI building), rc19 chunks (lint #13 contract + velocity backfill) **Validated lint score on Howard's existing session:** 25/33 → 27/33 (81.8%) after rc19 chunk

Status legend: –  MET — fully satisfies PRD –  PARTIAL — schema present, value needs source / operator action –  MISSING — not yet implemented –  IN CI — fix in flight –  OPERATOR — depends on recording-time human action –  CONTRACT — depends on customer signoff

§3.1 — recording.mp4 (video file, 9 PRD items)

#	PRD Requirement	Current State	Gap	ETA
1	Duration 5–6 min	Operator–controlled; recorder doesn’t enforce	👤 user records longer than 5 min next time	Now
2	Resolution 1920×1080	rc17.0.4+ locked at 1080p ✓	✅ MET	—
3	FPS 30 stable	rc17.0.4+ validated, real recording shows 29.97 ✓	✅ MET	—
4	Encoding H.264/H.265 CRF ≤ 23	rc17.0.4+ x264 CRF 22; AMD 780M falls back from HEVC	⚠️ H.264 ok per PRD but PRD prefers H.265; need GPU–specific HEVC config	rc19
5	Audio AAC continuous	OBS AAC 160kbps stereo configured ✓	⚠️ Howard’s session had real >2s silence gap (hardware issue)	Investigate minipc1 audio device
6	Input → frame latency ≤ 20ms	Not measured; <code>bin/measure_input_latency.py</code> exists but not run automatically	❌ <code>measure_input_latency</code> needs running before each release; not wired to CI	rc18.0.8
7	Fullscreen 1080p (system + game)	OBS records monitor, MC launched fullscreen	✅ MET	—
8	No UI/ inventory/ death/ perspective–switch	No enforcement; operator守则	👤 operator follows指南	Now
9	Per–game daily limit 240 clips	No cross–session counter	❌ backend tracking needed	rc20 backend

§3.1 Score: 4/9 ✅, 2/9 ⚠️, 1/9 ❌, 2/9 👤 = 44% strict / 67% with operator

§3.2 — `action_camera.json` (per-frame 20 fields)

Field	PRD	rc18.0.5 + rc19 chunks	Gap	ETA
<code>frame</code>	int continuous 0..N-1	recorder writes ✓	✓ MET	—
<code>time</code>	ISO ms format	recorder writes relative seconds	⚠ format mismatch — needs ISO conversion	rc19 next chunk
<code>fps</code>	30.0 explicit	metadata has fps_effective; per-frame fps missing	⚠ add to finalize	rc19 next chunk
<code>route_type</code>	∈ {1,2,3}	rc18.0.4 env- var default 1 ✓	⚠ operator must set per-session via env-vars	👤 per session
<code>mouse_x</code> / <code>mouse_y</code> (normalized 0–1)	Real position	rc17.3 BG fix; stuck at 0.5 if input pipeline broken	🔄 rc18.0.6 keyboard wake fix (CI); rc18.0.7 mouse hook install	rc18.0.7
<code>mouse_dx</code> / <code>mouse_dy</code> (per- frame delta)	Real delta	Same as above	🔄 rc18.0.6 / rc18.0.7	rc18.0.7
<code>keyCode</code> (pressed keys array)	Real key codes	rc18.0.6 fixes keyboard event pipeline (CI building)	🔄 IN CI	rc18.0.6 ~15 min
<code>camera_position</code> (x,y,z, m, LH coord)	Real position	rc19 chunk: backfilled from game_state ✓	✓ MET	—
<code>camera_rotation_oula</code> (Euler)	pitch [–90,90] yaw/roll [–180,180]	rc19 chunk: backfilled ✓	✓ MET	—
<code>camera_rotation_quaternion</code> xyzw,	q	≈1	Order + norm	rc19 chunk: backfilled + metadata declares “xyzw” ✓

Field	PRD	rc18.0.5 + rc19 chunks	Gap	ETA
<code>camera_Follow Offset</code>	First-person = 0	rc19 chunk: [0,0,0] ✓	✓ MET	—
<code>camera_intrinsics</code> <code>fx==fy</code>	Pinhole model	rc17.2.3+ <code>fx==fy=771.2</code> ✓	✓ MET	—
<code>camera_speed</code> (x,y,z, m/s per axis)	Real velocity	rc19 chunk: backfilled from <code>game_state velocity_x/y/z</code> ✓	✓ MET (NEW)	—
<code>player_position</code> (x,y,z, m, LH)	Real position	rc19 chunk: backfilled ✓	✓ MET	—
<code>player_rotation_oula</code>	Same as camera	rc19 chunk: backfilled ✓	✓ MET	—
<code>player_rotation_quaternion</code> xyzw	Same as camera	rc19 chunk: ✓	✓ MET	—
<code>player_speed</code> (x,y,z, m/s)	Real velocity	rc19 chunk: backfilled from <code>game_state</code> ✓	✓ MET (NEW)	—
<code>metric_scale</code>	1.0	recorder ✓	✓ MET	—
<code>input_modality</code>	“keyboard_mouse”	recorder ✓	✓ MET	—
<code>gamepad_buttons</code> / <code>gamepad*_stick_*</code>	Gamepad data	recorder reserves fields	✓ MET (null for kb+m sessions)	—

§3.2 Score: 16/20 ✓, 3/20 ⚠, 2/20 🔄 = 80% strict / 95% post-rc18.0.6

§3.3 — `gameinfo.xlsx` (single sheet, 14 fields)

Field	PRD	Current source	Gap	ETA
game_name	"Minecraft"	default ✓	✓ MET	—
game_version	"1.21.4" or similar	default ✓	✓ MET	—
platform	"Java Edition"	default ✓	✓ MET	—
scene_name	flat-overworld etc.	rc19 chunk: backfilled from game_state dimension ✓	✓ MET (NEW)	—
weather	clear/rain/thunder	env-var default "clear"	⚠ mc-mod doesn't emit weather; needs §3.3 Java patch	rc18.0.8 mc-mod
time_of_day	day/night/dawn/dusk	env-var default "day"	⚠ mc-mod doesn't emit time_of_day; needs §3.3 Java patch	rc18.0.8 mc-mod
character_name	DataPilot	env-var default ✓	✓ MET	—
character_class	survival/explorer	env-var default ✓	✓ MET	—
operator_id	vendor-001-op-A	env-var ✓	✓ MET (operator sets per-session)	—
recording_date	ISO date	metadata-derived ✓	✓ MET	—
total_frames	int from frames.jsonl	computed ✓	✓ MET	—
video_duration_sec	float from metadata	computed ✓	✓ MET	—
route_type	1/2/3	env-var per-session	⚠ operator sets	👤
notes	string	env-var optional	✓ MET (allowed to be empty)	—

§3.3 Score: 11/14 ✓, 3/14 ⚠ = 79% strict / 100% post-mc-mod weather/time emit

§3.4 — `depth/*.exr` (depth maps @ 6 fps)

Item	PRD	Current State	Gap	ETA
Sampling rate	6 fps (1800 frames / 5 min)	DISABLED in rc17.3.1 submodule	✗ depth EXR writer broken (rc17.4 cluster cv2 attempt captured desktop post-record at 1 Hz)	rc19 rewrite
Format	OpenEXR single-channel Z	N/A	✗	rc19
Dtype	float32	N/A	✗	rc19
Units	meters (linear along Z)	N/A	✗	rc19
Invalid pixels	0 (sky/transparent/clipped)	N/A	✗	rc19
Filename	<code>000000.exr</code> (t=0) timestamp-aligned	N/A	✗	rc19
Resolution	1920×1080 same as mp4	N/A	✗	rc19

§3.4 Score: 0/7  = 0% — entire deliverable blocked on rc19

Implementation plan: cv2-based mp4 re-decode + DepthAnything V2 Small (onnxruntime-directml) inference at 6 fps, output OpenEXR float32 Z-channel. Major Python+model work, ~1–2 day cycle including model bundling in installer.

§4 — Path diversity (per-batch)

Item	PRD	Current State	Gap	ETA
route_type distribution 50/25/25 (normal/special/loop)	Per-batch tracker	<code>bin/ batch_tracker.py</code> ✓ exists	✓ MET (operator runs it)	—
WASD balance	No direction starved	rc18.0.6 keyboard wake fix needed first	🔄 IN CI	rc18.0.6
No 1+ min stationary	Continuous input	Currently broken (input pipeline drops); rc18.0.6 fixes keyboard, rc18.0.7 fixes mouse	🔄	rc18.0.6/0.7
Path type coverage 1+2+3	Operator守则	per-session env-var	👤	—

§4 Score: 2/4 ✓, 2/4 🔄 = 50% strict / 100% post-rc18.0.7

§6 — Lint v3 acceptance (32 criteria, soon 33 with audio)

Howard's test session current score (after rc19 chunks): – ✓ 27 / 33 = 81.8% PASS – Failing: #2 (duration too short — operator), #15/#16/#24 (depth disabled — rc19), #31 (mouse pipeline — rc18.0.7), #38 (real audio dropout in this specific recording)

Projected post-rc18.0.6 + 5-min fresh recording: 28–29/33 Projected post-rc18.0.7 (mouse hook install): 29–30/33 Projected post-rc19 (depth + lint refinements): 32–33/33 100% = customer RFC-001 signoff (real-player path accepted)

Summary scorecard

PRD section	Strict %	Achievable with already-in-flight work	What blocks remaining
§3.1 mp4	44%	78% (after audio device check + latency measurement)	Latency wire-up (rc18.0.8)
§3.2 action_camera	80%	95% post-rc18.0.6/0.7	Mouse pipeline fix
§3.3 gameinfo	79%	100% post-mc-mod weather/time	mc-mod Java patch (rc18.0.8)
§3.4 depth	0%	0% until rc19	cv2 depth rewrite
§4 path diversity	50%	100% post-rc18.0.7	Mouse pipeline
§6 lint v3	82%	100% post-rc19 + signoff	Depth + RFC-001

Overall PRD strict compliance: ~55% **After rc18.0.6 lands** (~15 min from now): ~62% **After rc18.0.7 ships** (~1 hour Rust work): ~70% **After rc19 ships** (~1–2 days for depth): ~92% **Plus RFC-001 customer signoff: 100% contract delivery**

Outstanding work tickets

🔥 In flight (CI)

- **rc18.0.6** building now (~15 min to URL): keyboard hook `PostThreadMessageW` wake fix → unblocks #27 (real keyboard events in `inputs.jsonl`) + partial #31

🔧 Engineering (cluster or human push needed)

- **rc18.0.7:** `SetWindowsHookExW(WH_MOUSE_LL, mouse_ll_proc, ...)` install call — `mouse_ll_proc` defined but never installed; mouse capture 100% dead on tier-3 (AMD/Intel iGPU) systems
- **rc18.0.8:** (a) mc-mod Java patch to emit `weather` + `time_of_day` per tick; (b) `measure_input_latency.py` wire into post-record finalize for §3.1 latency check
- **rc19:** depth EXR cv2 mp4-decoder + DepthAnything V2 Small inference at 6 fps + OpenEXR float32 Z-channel writer
- **rc19:** H.265 GPU detection — try AMD AMF / NVIDIA NVENC / Intel QSV before x264 fallback

- **rc19:** lint v3 criteria 38–42 wiring (drafts in `oyster-audit/drafts/`)
- **rc20:** backend cross-session counters (240 clips/game/day, 30-min/map limits)

Customer

- **RFC-001 signoff:** real-player vs headless-bot recording path acceptance
(`docs/RFC-001-real-player-vs-bot.md`)

User actions

- Install rc18.0.6 installer when CI lands (~15 min)
 - Record via Start Menu → “Launch Minecraft (Recorded)” (NOT Mojang Launcher)
 - Record **5+ minutes** per session (PRD §3.1 duration requirement)
 - Set operator env-vars per session: `OYSTER_OPERATOR_ID` , `OYSTER_CHARACTER_NAME` , `OYSTER_ROUTE_TYPE` , etc.
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What’s NOT a problem

These items HAVE caused confusion in earlier reviews but are actually correct:

- **action_camera.json 29 fields instead of 20** — extra fields are PRD-allowed extensions (`gamepad_*`, `frame_index` alias, `input_modality`). PRD §3.2 lists 20 REQUIRED; ours has all 20 + 9 informative extras.
 - **H.264 instead of H.265 on AMD 780M** — PRD §3.1 allows either; the GPU just doesn’t have HEVC. Same data quality.
 - **Lint #13 heuristic-FAIL on rc17.x sessions** — heuristic flaw on game data with large rotation; rc19 contract check via `metadata.json` fixes this for all future sessions.
 - **action_camera quaternion=null on rc18.0.3 sessions** — recorder wrote file before mc-mod `game_state.jsonl` was synced; rc18.0.5’s `finalize_session.py` backfills post-record, runs automatically via `prd_acceptance.py` .
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Reference paths

- Lint runner: `bin/lint_v3_prd_grounded.py` — 33 criteria
- Acceptance harness: `bin/prd_acceptance.py` — runs all `prd_test_*.py` + lint + outputs `PRD-ACCEPTANCE-REPORT.md`

- Finalize step: `bin/finalize_session.py` — sync game_state + backfill quaternion/velocity/scene + generate gameinfo + generate audio_check
- Operator metadata form: `bin/oyster_launcher_form.py` — first-launch + per-session
- Customer RFC: `docs/RFC-001-real-player-vs-bot.md`
- Audit reports: `oyster-audit/` directory